

भारतीय मानक
ऊष्मारोधी द्रव — विद्युतीय प्रयोजनार्थ अनुप्रयुक्त
सिंथेटिक कार्बनिक इस्टर की विशिष्टियाँ

Indian Standard

INSULATING LIQUIDS — SPECIFICATIONS FOR
UNUSED SYNTHETIC ORGANIC ESTERS FOR
ELECTRICAL PURPOSES

ICS 29.040

© BIS 2013

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

NATIONAL FOREWORD

This Indian Standard which is identical with IEC 61099 : 2010 'Insulating liquids — Specifications for unused synthetic organic esters for electrical purposes' issued by the International Electrotechnical Commission (IEC) was adopted by the Bureau of Indian Standards on the recommendation of the Fluids for Electrotechnical Applications Sectional Committee and approval of the Electrotechnical Division Council.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
- b) Comma (,) has been used as a decimal marker, while in Indian Standards the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their respective places are listed below along with their degree of equivalence for the editions indicated:

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
IEC 60156 Insulating liquids — Determination of the breakdown voltage at power frequency — Test method	IS 6792 : 1992 Method for determination of electric strength of insulating oils (<i>first revision</i>)	Technically Equivalent
IEC 60475 Method of sampling liquid dielectrics	IS 6855 : 2003 Method of sampling for liquid dielectrics (<i>first revision</i>)	do
IEC 60628 : 1985 Gassing of insulating liquids under electrical stress and ionization	IS 12475 (Part 1) : 1988 Gassing of insulating liquid under electric stress and ionization: Part 1 Method of determination of gassing rate under hydrogen atmosphere	do
IEC 60814 Insulating liquids — Oil-impregnated paper and pressboard — Determination of water by automatic coulometric Karl Fischer titration	IS 13567 : 1992 Determination of water in insulating liquids and in oil-impregnated paper and pressboard by automatic coulometric Karl Fischer titration — Method of test	do
IEC 61039 Classification of insulating liquids	IS 13503 : 2013 Classification of insulating liquids	Identical to IEC 61039 : 2008
IEC 61620 Insulating liquids — Determination of the dielectric dissipation factor by measurement of the conductance and capacitance — Test method	IS 16086 : 2013 Insulating liquids — Determination of the dielectric dissipation factor by measurement of the conductance and capacitance — Test method	Identical to IEC 61620 : 1998
ISO 2592 Determination of flash and fire points — Cleveland open cup method	IS 1448 [P : 69] : 2004 Methods of test for petroleum and its products: [P : 69] Flash and fire point by cleveland (open) cup	Technically Equivalent

(Continued on third cover)

INTRODUCTION

Health and safety

This International Standard does not purport to address all the safety problems associated with its use. It is the responsibility of the user of the standard to establish appropriate health and safety practices and determine the applicability of regulatory limitations prior to use.

Unused synthetic esters which are the subject of this standard should be handled with due care with regard to personal hygiene. Direct contact with eyes may cause slight irritation. In the case of eye contact, irrigation with copious quantities of clean running water should be carried out and medical attention sought.

Some of the tests specified in this standard involve the use of processes that could lead to a hazardous situation. Attention is drawn to the relevant standard for guidance.

Environment

The disposal of synthetic esters, chemicals and sample containers mentioned in this standard should be carried out in accordance with local regulations with regard to their environmental impact. Precautions should be taken to prevent the release of synthetic esters into the environment.

Indian Standard

INSULATING LIQUIDS — SPECIFICATIONS FOR UNUSED SYNTHETIC ORGANIC ESTERS FOR ELECTRICAL PURPOSES

1 Scope

This International Standard covers the specification and test methods for unused synthetic organic esters.

It applies to synthetic organic esters, delivered to the agreed point and time of delivery intended, for use in transformers, switchgear and similar related equipment in which synthetic organic esters are required as an insulant and for heat transfer. These unused synthetic organic esters are obtained by chemical processing and physical treatments of fatty acids and polyols.

NOTE Maintenance of synthetic organic esters in equipment is covered in a separate standard (IEC 61203).

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60156, *Insulating liquids – Determination of the breakdown voltage at power frequency – Test method*.

IEC 60247, *Insulating liquids – Measurement of relative permittivity, dielectric dissipation factor ($\tan \delta$) and d.c. resistivity*

IEC 60475, *Method of sampling liquid dielectrics*

IEC 60628:1985, *Gassing of insulating liquids under electrical stress and ionization*

IEC 60814, *Insulating liquids – Oil-impregnated paper and pressboard – Determination of water by automatic coulometric Karl Fischer titration*

IEC 61039, *Classification of insulating liquids*

IEC 61125:1992, *Unused hydrocarbon-based insulating liquids – Test methods for evaluating the oxidation stability*

IEC 61620, *Insulating liquids – Determination of the dielectric dissipation factor by measurement of the conductance and capacitance – Test method*

IEC 62021-1, *Insulating liquids – Determination of acidity – Part 1: Automatic potentiometric titration*

IEC 62021-2, *Insulating liquids – Determination of acidity – Part 2: Colourimetric titration*

ISO 2211, *Liquid chemical products – Measurement of colour in Hazen units (platinum-cobalt scale)*

ISO 2592, *Determination of flash and fire-points – Cleveland open cup method*

ISO 2719, *Determination of flash-point – Pensky Martens closed cup method*

ISO 3016, *Petroleum products – Determination of pour-point*

ISO 3104, *Petroleum products – Transparent and opaque liquids – Determination of kinematic viscosity and calculation of dynamic viscosity*

ISO 3675, *Crude petroleum and liquid petroleum products – Laboratory determination of density – Hydrometer method*

ISO 12185, *Crude petroleum and petroleum products – Determination of density – Oscillating U-tube method*

OECD 301:1992, *OECD guideline for testing of chemicals – Ready biodegradability*

3 Terms and definitions

For the purposes of this document the following terms and definitions apply.

3.1

unused synthetic organic esters

liquid based on synthetic organic esters as delivered by the supplier

3.2

additive

suitable chemical substance/s which is/are deliberately added to a synthetic organic ester in order to improve certain characteristics, e.g. pour point, viscosity, foaming, oxidation stability

NOTE If additives are used, these should be mentioned by the supplier to the user, if requested. This may involve issues of confidentiality in an agreement. Additives, if used, need to comply with local regulations.

4 General remarks

The unused synthetic organic esters to which this standard applies are liquids which contain only carbon, hydrogen and oxygen. They are prepared from mono- or polyhydric alcohols and mono- or polybasic aliphatic or aromatic acids. Commercial products may be based on single esters or a mixture of esters and may contain oxidation inhibitors and other additives.

Unused synthetic organic esters selected for use in transformers have high flash and fire points and are therefore relatively difficult to ignite. They may, however, have slightly higher viscosities than mineral insulating oils.

Unused synthetic organic esters are more hygroscopic than mineral oils and this aspect will need to be considered in their use and maintenance.

NOTE Such a liquid, by definition, has not been used in, nor been in contact with, electrical equipment or other equipment not required for manufacture, storage or transport. The manufacturers and suppliers of unused synthetic organic esters will have taken all reasonable precautions to ensure that the liquid is not contaminated with polychlorinated biphenyls or terphenyls (PCB, PCT); used, reclaimed or dechlorinated oils; or other contaminants.

5 Health, safety and environment (HSE) properties (liquid properties related to safe handling and environment protection)

5.1 Flash point and fire point

The safe operation of electrical equipment requires an adequately high fire point that is measured in accordance with ISO 2592 (Cleveland open cup). Flash point is measured with ISO 2719 (Pensky-Martens closed cup procedure).

5.2 Biodegradation

Unused synthetic organic esters as supplied in electrical applications are of interest because of their higher fire points and better environmental compatibility, relative to petroleum derived insulating mineral oil. Specific tests need to be undertaken to demonstrate biodegradability of these liquids. Tests include OECD 301 / OPPTS 835.3110 (US EPA). The biodegradability of unused synthetic ester fluids can be classified using OECD 301 according to the IEC 61039.

5.3 Toxicity

Unused synthetic organic esters are considered non-toxic and suppliers shall supply assays that define the product as non-toxic.

NOTE Toxicity of unused synthetic organic ester can be assessed with test methods such as modified Ames assay or other suitable internationally recognized assays.

6 Classification, identification, general delivery requirements, sampling, packaging and disposal

6.1 Classification

For the purpose of this standard, unused synthetic ester insulating liquids are classified into a single class: less flammable synthetic organic ester liquids.

6.2 Identification and general delivery requirements, sampling, packaging and disposal

Requirements shall be as follows:

- a) Unused synthetic organic esters are normally delivered in bulk, in rail tank cars, tank containers or packed in drums or intermediate bulk containers. These shall be clean and suitable for this purpose to avoid any contamination.
- b) Liquid drums and sample containers shall carry at least the following markings:
 - supplier's designation;
 - classification;
 - liquid net weight.
- c) Unused synthetic organic ester delivery shall be accompanied by a document from the supplier specifying at least: supplier's designation, liquid classification and quality certificate. At the purchaser's request, the supplier shall indicate the presence of any additive.

6.3 Storage

Storage should preferably be indoors and in suitable closed containers to prevent ingress of contamination. In the event of accidental contamination by water and/or solid particles during storage, the liquid can normally be restored to an acceptable quality by vacuum treatment followed by filtration or by using solid drying agents such as molecular sieves, followed by filtration.

6.4 Representative sampling

Representative sampling shall be carried out in accordance with the procedure described in IEC 60475.

6.5 Disposal and spillages

Synthetic organic esters should be disposed of in accordance with local regulations. Spillages should be cleaned according to the directions given in the materials safety data sheet.

7 Properties of synthetic organic esters

NOTE Characteristics are listed in specification Sheet 1.

7.1 Physical properties, significance and test methods

7.1.1 Colour

Colour shall be measured according to ISO 2211.

7.1.2 Appearance

A visual inspection of synthetic ester liquids (oil sample in transmitted light under a thickness of approximately 10 cm and at ambient temperature) indicates the presence of visible contaminants, free water and suspended matter.

7.1.3 Density

Density of the liquid shall be measured in accordance with ISO 3675 or ISO 12185.

7.1.4 Kinematic viscosity

Kinematic viscosity shall be measured according to ISO 3104.

NOTE The cold start behaviour of synthetic organic esters may be different from mineral oil.

7.1.5 Flash point and fire point

The safe operation of electrical equipment requires an adequately high flash and fire point. The flash point is measured by ISO 2719 (Pensky-Martens closed cup procedure), and the fire point is measured in accordance with ISO 2592 (Cleveland open cup).

7.1.6 Pour point

The pour point shall be determined according to ISO 3016.

7.1.7 Crystallization

Crystallisation shall be determined according to the test method description in Annex A.

7.1.8 Water content

Water content shall be measured in accordance with IEC 60814.

7.1.9 Acidity

The acidity should be determined in accordance with IEC 62021-1 or IEC 62021-2.

NOTE Precision data reported in these methods do not apply to synthetic esters.

7.1.10 Oxidation stability

Oxidation stability of synthetic organic esters can be assessed with IEC 61125:1992, Method C for 164 h. There is also the option to test at 500 h to allow for the extra stability of synthetic organic esters. This extra test is optional with no requirement in this standard.

8 Electrical properties

8.1 Breakdown voltage

Breakdown voltage of unused synthetic esters insulating liquid shall be measured in accordance with IEC 60156. However, because of the different properties of synthetic esters as compared to mineral oil, an extended initial set up time is required.

NOTE To include the wait time for the extra viscosity of synthetic ester, some modification of the method may be needed. This is an extension of the time between pouring into the test cell and testing to allow gas bubbles to come out of the fluid. It is recommended that a wait period of at least 1 h between loading the cell and the first breakdown arc is recommended. Also to ensure that there are no visible bubbles in the liquid.

8.2 Dielectric dissipation factor, permittivity and d.c. resistivity

The dielectric dissipation factor (DDF) is a measure for dielectric losses caused by the liquid. Increased DDF can indicate contamination of the liquid by moisture, particles or soluble polar contaminants or poor refining quality. DDF shall be measured in accordance with IEC 60247 or IEC 61620 both methods at 90 °C. In case of dispute, IEC 60247 at 90 °C should be used.

8.3 Gassing tendency

Gassing tendency shall be measured according to Method A of IEC 60628:1985 but there is no requirement in this standard.

9 Specifications for synthetic organic esters used in transformers

When tested in accordance with the methods specified in this standard, the properties of the unused liquid shall comply with the requirements given in specification sheet 1.

Specification sheet 1 –
Transformer ester – Type T1

This type of ester is prepared from a polyhydric alcohol (polyol) and a mixture of monocarboxylic acids with suitable stabilizing additives as required to meet the properties specified below.

Property	Test method	Permissible values
Physical		
Colour	ISO 2211	Max. 200 Hazen
Appearance	Visual	Clear, free from water and suspended matter and sediment
Density at 20 °C (kg/dm ³)	ISO 3675 or ISO 12185	Max. 1 000
Kinematic viscosity at 40 °C (mm ² /s) at –20 °C (mm ² /s)	ISO 3104	Max. 35 Max. 3 000
Flash-point (°C)	ISO 2719	Min. 250
Fire-point (°C)	ISO 2592	Min. 300
Pour-point (°C)	ISO 3016	Max. –45
Chemical		
Water content (mg/kg)	IEC 60814	Max. 200 ^a
Acidity (mg KOH/g)	IEC 62021-1 or IEC 62021- 2	Max. 0,03
Oxidation stability ^b	IEC 61125, Method C	
Test duration 164 h		
Total acidity (mg KOH/g)		Max. 0,3
Total sludge (% mass)		Max. 0,01
Electrical		
Breakdown voltage (kV)	IEC 60156	Min. 45 ^a (See Clause 8)
Dielectric dissipation factor, tan δ at 90 °C and 50 Hz	IEC 60247 or IEC 61620	Max. 0,03 ^{a, c}
DC resistivity at 90 °C (G Ω × m)	IEC 60247	Min. 2
^a For untreated liquid, as received. ^b Oxidation stability (IEC 61125, Method C) at 500 h is an optional extra test. No requirement in this standard. ^c For frequencies (f (Hz)) in the range of 48 Hz to 62 Hz, convert values as follows: $\tan \delta [f (50 \text{ Hz})] = \frac{f (\text{Hz})}{50} \times \tan \delta [f (\text{Hz})]$		

Annex A (informative)

Determination of crystallization

A.1 General comment

Determining the likelihood of crystal formation in synthetic organic esters could be a requirement for type testing.

A.2 Outline of the method

This method describes a procedure for assessing the tendency of synthetic organic esters to crystallize under certain specific conditions.

A.3 Apparatus

The apparatus shall consist of

- a) crystallizing dish of adequate capacity (150 ml to 250 ml),
- b) a refrigerator able to maintain a temperature of $-25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$.

A.4 Procedure

Proceed as follows:

- a) Pour approximately 100 ml of the synthetic organic ester to be tested into the crystallizing dish.
- b) Store the crystallizing dish filled with the synthetic ester in the refrigerator for 16 h maintained at $-25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$.
- c) At the end of the test period examine the liquid sample visually in order to detect the presence of crystals.

Annex B (informative)

Specifications for synthetic organic esters used in capacitors

This specification concerning synthetic organic esters used in capacitors is based on the previous version of this standard, and is only included here for information.

When tested in accordance with the methods specified in this standard, the properties of the unused synthetic organic esters shall comply with the requirements given in specification sheet 2.

Specification sheet 2 –

Capacitor ester – Type C1

This type of ester usually consists entirely of di-2-ethylhexyl orthophthalate.

Property	Test method	Permissible values
Physical		
– Colour	ISO 2211	Max. 200
– Appearance	Visual	Clear, free from suspended matter and sediment
– Density at 20 °C (kg/dm ³)	ISO 3675 or ISO 12185	Max. 0,990
– Viscosity at 40 °C (mm ² /s)	ISO 3104	Max. 30
– Flash-point (°C)	ISO 2719	Min. 190
– Fire-point (°C)	ISO 2592	Min. 220
– Pour-point (°C)	ISO 3016	Max. –39
Chemical		
– Water content (mg/kg)	IEC 60814	Max. 200 ^a
– Neutralization value (mg KOH/g)	IEC 62021- 1 or IEC 62021- 2	Max. 0,03
Electrical		
– Breakdown voltage (kV)	IEC 60156	Min. 50 ^a
– Dielectric dissipation factor, tan δ at 90 °C and 50 Hz	IEC 60247 or IEC 61620	Max. 0,03 ^{a, b}
– Permittivity at 90 °C and 50 Hz	IEC 60247	4,2 to 4,4
^a For untreated liquid, as received. ^b For frequencies (f (Hz)) in the range of 48 Hz to 62 Hz, convert values as follows: $\tan \delta [f (50 \text{ Hz})] = \frac{f (\text{Hz})}{50} \times \tan \delta [f (\text{Hz})]$		

Bibliography

IEC 61203, *Synthetic organic esters for electrical purposes – Guide for maintenance of transformer esters in equipment*

ISO 5661, *Petroleum products – Hydrocarbon liquids – Determination of refractive index*

(Continued from second cover)

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
ISO 2719 Determination of flash-point — Pensky Martens closed cup method	IS 1448 [P : 66] : 2004 Methods of test for petroleum and its products: [P : 66] Flash point (open) and fire point by Pensky-Martens apparatus	Technically Equivalent
ISO 3016 Petroleum products — Determination of pour-point	IS 1448 [P : 10] Methods of test for petroleum and its products: [P : 10] Determination of pour point (<i>second revision</i>) (<i>under print</i>)	do
ISO 3104 Petroleum products — Transparent and opaque liquids — Determination of kinematic viscosity and calculation of dynamic viscosity	IS 1448 [P : 25] Methods of test for petroleum and its products: [P : 25] Transparent and opaque liquids — Determination of kinematic viscosity and calculation of dynamic viscosity (<i>second revision</i>) (<i>under print</i>)	do
ISO 3675 Crude petroleum and liquid petroleum products — Laboratory determination of density — Hydrometer method	IS 1448 [P : 16] Methods of test for petroleum and its products: [P : 16] Lab method for density — Hydrometer method (<i>fourth revision</i>) (<i>under print</i>)	do

The technical committee has reviewed the provisions of the following International Standards/Other Publications referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

<i>International Standard/ Other Publication</i>	<i>Title</i>
IEC 60247	Insulating liquids — Measurement of relative permittivity, dielectric dissipation factor ($\tan \delta$) and d.c. resistivity
IEC 61125 : 1992	Unused hydrocarbon-based insulating liquids — Test methods for evaluating the oxidation stability
IEC 62021-1	Insulating liquids — Determination of acidity — Part 1: Automatic potentiometric titration
IEC 62021-2	Insulating liquids — Determination of acidity — Part 2: Colourimetric titration
ISO 2211	Liquid chemical products — Measurement of colour in Hazen units (platinum-cobalt scale)
ISO 12185	Crude petroleum and petroleum products — Determination of density — Oscillating U-tube method
OECD 301 : 1992	OECD guideline for testing of chemicals — Ready biodegradability

Only the English language text has been retained while adopting it in this Indian Standard and as such the page numbers given here are not the same as in the IEC Publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act*, 1986 to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright

BIS has the copyright of all its publications. No part of these publications may be reproduced in any form without the prior permission in writing of BIS. This does not preclude the free use, in course of implementing the standard, of necessary details, such as symbols and sizes, type or grade designations. Enquiries relating to copyright be addressed to the Director (Publications), BIS.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards: Monthly Additions'.

This Indian Standard has been developed from Doc No.: ETD 03 (6352).

Amendments Issued Since Publication

Amendment No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

Telephones: 2323 0131, 2323 3375, 2323 9402

Website: www.bis.org.in

Regional Offices:

Telephones

Central	: Manak Bhavan, 9 Bahadur Shah Zafar Marg NEW DELHI 110002	{ 2323 7617 2323 3841
Eastern	: 1/14, C.I.T. Scheme VII M, V.I.P. Road, Kankurgachi KOLKATA 700054	{ 2337 8499, 2337 8561 2337 8626, 2337 9120
Northern	: SCO 335-336, Sector 34-A, CHANDIGARH 160022	{ 260 3843 260 9285
Southern	: C.I.T. Campus, IV Cross Road, CHENNAI 600113	{ 2254 1216, 2254 1442 2254 2519, 2254 2315
Western	: Manakalaya, E9 MIDC, Marol, Andheri (East) MUMBAI 400093	{ 2832 9295, 2832 7858 2832 7891, 2832 7892

Branches: AHMEDABAD. BANGALORE. BHOPAL. BHUBANESHWAR. COIMBATORE. DEHRADUN. FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KANPUR. LUCKNOW. NAGPUR. PARWANOO. PATNA. PUNE. RAJKOT. THIRUVANATHAPURAM. VISAKHAPATNAM.